Java String

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What is String in Java?

- Generally, String is a sequence of characters. But in Java, string is an object that represents a sequence of characters. The java.lang.String class is used to create a string object.

How to create a string object?

- There are two ways to create String object:

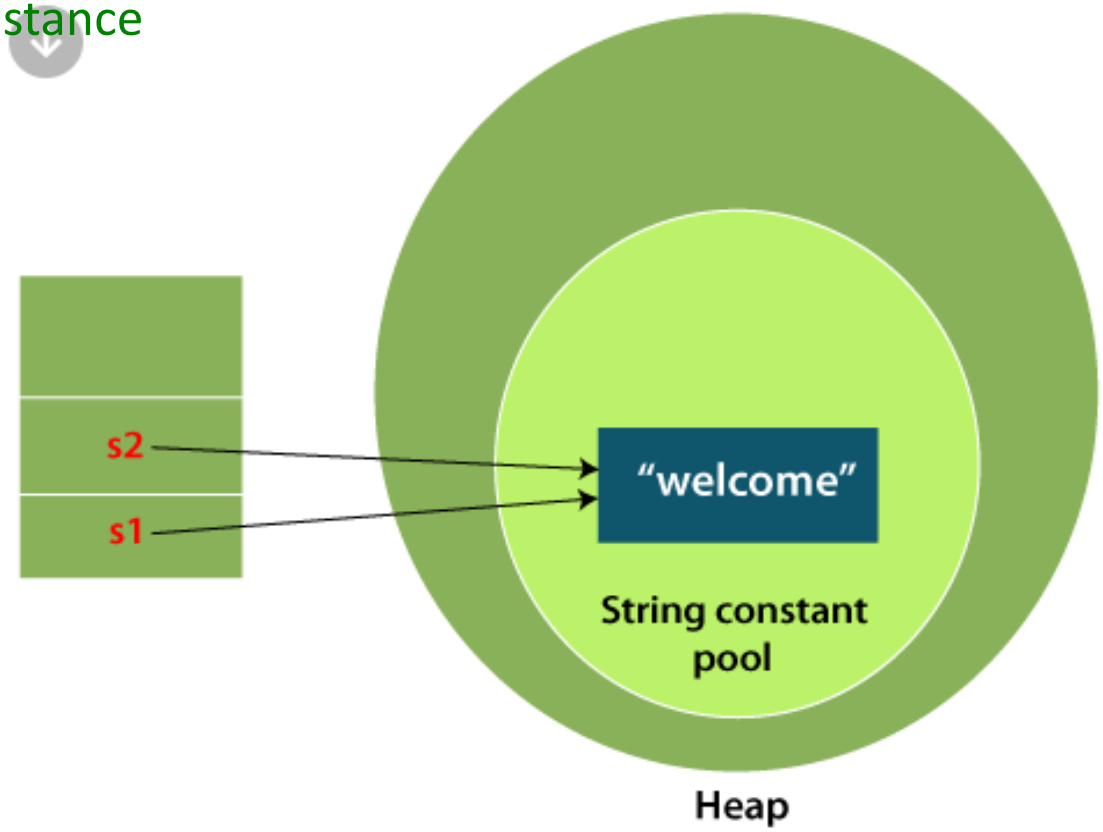
1. By string literal
2. By new keyword

String Literal

- Java String literal is created by using double quotes. For Example:
- String s="welcome";
- Each time you create a string literal, the JVM checks the "string constant pool" first. If the string already exists in the pool, a reference to the pooled instance is returned. If the string doesn't exist in the pool, a new string instance is created and placed in the pool. For example:

String Literal … Cont

1. String s1="Welcome";
2. String s2="Welcome";//It doesn't create a new instance



- In the above example, only one object will be created. Firstly, JVM will not find any string object with the value "Welcome" in string constant pool that is why it will create a new object. After that it will find the string with the value "Welcome" in the pool, it will not create a new object but will return the reference to the same instance.

By new keyword

```
String s=new String("Welcome");//creates two objects and one referen
ce variable
```

In such case, JVM will create a new string object in normal (non-pool) heap memory, and the literal "Welcome" will be placed in the string constant pool. The variable s will refer to the object in a heap (non-pool).

Java String Example

```
public class StringExample{
public static void main(String args[]){
String s1="java";//creating string by Java string literal
char ch[]={'s','t','r','i','n','g','s'};
String s2=new String(ch);//converting char array to string
String s3=new String("example");//creating Java string by new keyword

System.out.println(s1);
System.out.println(s2);
System.out.println(s3);
 }}
```

Java String class methods

1- Java String charAt()

```
char Variable =String.charAt(int index)
```

Example :

FileName: CharAtExample.java

```
public class CharAtExample{
public static void main(String args[]){
String name="javatpoint";
char ch=name.charAt(4);//returns the char value at the 4th index
System.out.println(ch);
}}
```

Output: t

2- Java String length()

Return length of string

- String.length()

Ex:FileName: LengthExample.java

```
public class LengthExample{
public static void main(String args[]){
String s1="javatpoint";
String s2="python";
System.out.println("string length is: "+s1.length());//10 is the length of ja
vatpoint string
System.out.println("string length is: "+s2.length());//6 is the length of pyt
hon string
}}
```

3- Java String substring() method example

```
public class SubstringExample{
public static void main(String args[]){
String s1="javatpoint";
System.out.println(s1.substring(2,4));//returns va// from 2 to 4-1 index
System.out.println(s1.substring(2));//returns vatpoint // from 2 to end
}}
```

4.Java String contains()

- The **Java String class contains()** method searches the sequence of characters in this string. It returns *true* if the sequence of char values is found in this string otherwise returns *false*.

- **FileName:** ContainsExample.java

```
class ContainsExample{
public static void main(String args[]){
String name="what do you know about me";
System.out.println(name.contains("do you know"));
System.out.println(name.contains("about"));
System.out.println(name.contains("hello"));
}}
```

Java String equals()

- The **Java String class equals()** method compares the two given strings based on the content of the string. If any character is not matched, it returns false. If all characters are matched, it returns true.

  The String equals() method overrides the equals() method of the Object class.

- **FileName:** EqualsExample.java

```
public class EqualsExample{
public static void main(String args[]){
String s1="javatpoint";
String s2="javatpoint";
String s3="JAVATPOINT";
String s4="python";
System.out.println(s1.equals(s2));//true because content and case is same
System.out.println(s1.equals(s3));//false because case is not same
System.out.println(s1.equals(s4));//false because content is not same
System.out.println(s1. equalsIgnoreCase(s3));//True because use IgnoreCase
}}
```

Java String isEmpty()

```
public class IsEmptyExample{
public static void main(String args[]){
String s1="";
String s2="javatpoint";

System.out.println(s1.isEmpty());
System.out.println(s2.isEmpty());
}}
```

Java String split()

- The **java string split()** method splits this string against given regular expression and returns a string array.

Ex

```
public class Split Example{
public static void main(String args[]){
String s1="java string split method by javatpoint";
String[] words=s1.split("\\s");//splits the string based on whitespace
//using java foreach loop to print elements of string array
for(String w:words){
System.out.println(w);
}
}}
```

Java String toLowerCase()

- The **java string toLowerCase()** method returns the string in lowercase letter. In other words, it converts all characters of the string into lower case letter.
- Java String toLowerCase() method example

```
public class StringLowerExample{
public static void main(String args[]){
String s1="JAVATPOINT HELLO stRIng";
String s1lower=s1.toLowerCase();
System.out.println(s1lower);
}}
```

- Java String toUpperCase()

The **java string toUpperCase()** method returns the string in uppercase letter. In other words, it converts all characters of the string into upper case letter.

- Java String toUpperCase() method example

```
public class StringUpperExample{
public static void main(String args[]){
String s1="hello string";
String s1upper=s1.toUpperCase();
System.out.println(s1upper);
}}
```

Trim Function

The **Java String class trim()** method eliminates leading and trailing spaces. The Unicode value of space character is '\u0020'. The trim() method in Java string checks this Unicode value before and after the string, if it exists then the method removes the spaces and returns the omitted string.

```
public class StringTrimExample2 {
   public static void main(String[] args) {
      String s1 ="  hello java string   ";
      System.out.println(s1.length());
      System.out.println(s1); //Without trim()
      String tr = s1.trim();
      System.out.println(tr.length());
      System.out.println(tr); //With trim()
   }
}
```

ex1: use suitable method of string in this code

```
public class Student {
public static void main(String[] args) {
String city = "Delhi";
if(city == "Meerut") { System.out.println("city is meerut"); }
else if (city == "Noida") { System.out.println("city is noida"); }
else if(city == "Agra") { System.out.println("city is agra"); }
else { System.out.println(city);
  }
    }
}
```

Output:

Delhi

ex2: read string s1=“i am from iraq” then convert first letter from Iraq to capital later

```
public class JavaString2 {
public static void main(String[] args) {
String s1="i am from iraq";
String[] words = s1.split("\\s");
for (int i = 0; i < words.length; i++) {
if (words[i].equalsIgnoreCase("iraq")) {

words[i] = words[i].substring(0, 1).toUpperCase() + words[i].substring(1);
}
   }
for(String w:words)
    System.out.print(w+" ");
   }}
```